Subject

Sun Mon Tue Wed Thu Fri Sat

Task 5

Date:

Place

For k possible outcomes,

$P = p_1, p_2 \dots p_k$

↳ target

$Q = q_1, q_2 \dots q_k$

↳ model

entropy is the uncertainty in target distribution

↳ $H(P) = -\sum_{i=1}^{k} p_i \ln p_i$

cross-entropy measures negative log probability assigned by Q to outcomes drawn from P.

↳ $H(P,Q) = -\sum_{i=1}^{k} p_i \ln q_i$

Now, KL-divergence

↳ $D_{KL}(P \| Q) = \sum_{i=1}^{k} p_i \ln \frac{p_i}{q_i}$

$= \sum_{i=1}^{k} p_i (\ln p_i - \ln q_i)$

$= \sum_{i=1}^{k} p_i \ln p_i - \sum_{i=1}^{k} p_i \ln q_i$

$= -H(P) + H(P,Q)$

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So, $H(P,Q) = H(P) + D_{KL}(P||Q)$

when P is fixed, so is H(P). That means it will always return 0 as gradient returns rate of change and will always be 0. for constants.

Therefore, minimizing H(P,Q) is the same as minimizing $D_{KL}(P||Q)$ when P is fixed.

Why cross-entropy is standard for classification

↳ classification is always one hot encoded problem. So, P_i will either be 0 or 1. Therefore for the particular class, In $P_i = 0$, and for all other, $P_i = 0$. This together makes $H(P) = 0$.

For $H(P,Q) \rightarrow$ log decreases only when more probability is assigned to correct class, otherwise not. [as $P_i = 0$ for others]. This makes it suitable to use cross-entropy as standard for classification as tweaking values of other classes doesn't affect the log.